

March 2026

Global nickel strategic planning outlook Q1 2026



Executive summary

For most of last year nickel's flat to declining prices on the exchanges reflected the true condition of the market: one of significant excess – with the promise of yet more supply to come – and by disappointing demand in two of the metal's leading consuming markets. That imbalance was rendered irrelevant mid-December when Indonesia's "new rules" came into view, begging the question: would the world's leading nickel supplier be able to produce anything at all in 2026? Of course, the reality was far less dramatic, as such manufactured conditions often end up being so. But it was a "shock" to the nickel system just as significant as the current Middle East crisis is to a range of commodities beyond just oil.

Opinions on how long the latest conflict will endure vary, but already one impact on nickel producers requiring sulphur as an energy source is making itself known. That means Indonesia's high pressure acid leach (HPAL) facilities, which have a high dependency on sulphur shipped from the Middle East. Tie the rising cost and ultimate threat to availability of sulphur with the nagging doubts behind the ability of Indonesia's miners to extract the required amounts and types of ore and it is quite easy to anticipate a swift reversal in currently lagging nickel quotes on the LME and SHFE.

But we're not quite there yet in 2026. Where we find ourselves is staring at another two years of market surplus having already been through four. Not so long ago the world was touting electrification as the driving force behind high prices and offering a challenge to finding enough nickel. Sino-Indonesian mine-to-refinery investment has seen to that – western companies have missed that particular (electric?) bus. And yet just as those upstream investments crank out the metal, Chinese downstream consumers are proving that once considered "inferior" technology doesn't need it. Lithium-iron-phosphate chemistries have made the breakthroughs required to question the need for so much nickel in the world's largest battery market – and beyond.

Quite what Indonesian nickel operations will do with all that nickel is perhaps best answered by where we have seen it go so far: obtain approval to put it into LME depots. Indonesian cathode has become a financial reward to be deployed elsewhere for another time, but the metal is certainly no longer the product of admiration that it was of yore. Not even Chinese stainless melters are routinely using it anymore, less so dissolvers for the sulphate trade where risk and innovation led them to use lower-cost intermediates as standard.

And if these aren't sufficient warnings of what's to come for a western nickel sector stuck in its own legacy approach of working to a 15-year timeline for new mine or product development, consider what we see taking place in China now. Producers of cut-price commodity stainless grades, such as 304, are now moving up the value chain to 316 and are overcoming the technological barriers behind no-nickel ferritics. And they also have more high-nickel alloys and superalloys in their sights. Sure, these materials are intended for China's own commercial, military and space jet-engine programmes. But over the next decade we expect to find their products entering the wider world and competing against those western stainless companies which only recently moved aggressively into the high-nickel alloys business via acquisition – precisely because those lower-cost stainless exports from China had "eaten their lunch". Will revised protectionist de-carbonisation tariffs such as CBAM, this year coming into force on stainless exports to Europe, be required to shore up "home" industries once again? Perhaps the nascent programmes being fleshed out in Canada, the USA and Australia to protect and encourage domestic nickel mining industries is the only way to go after all.

Looking beyond these near-term challenges (and opportunities) of a burdened nickel market, we do see structural re-balancing by the end of this decade, and with it the prospect of higher prices. By 2031 we should have shaken off the malaise of US\$17,000/t prices (US\$7.70/lb) and entered the realms of US\$20,000/t (US\$9.10/lb, in US\$2026 terms) prices.

And by 2032 we will need nickel from new resources – around 100 kt or so, but growing to 1.3 Mt by 2040. The promise of annual average nickel prices aligning with our long-term price of US\$19,600/t (US\$8.90/b) should help incentivise such projects.

Market outlook and price

Medium-term forecasts

Almost throughout 2025, nickel markets and prices could have been considered as “forgettable”, with very few positives to say from month to month. Growth in supply surpassed that of demand threefold, resulting in ever-accumulating surpluses which inevitably took their toll on prices, with the first-half average cash price on the LME of US\$15,366/t (US\$6.97/lb) falling to US\$14,967/t (US\$6.79/lb) during H2. But then, right at the death during the latter days of trading in December, prices responded to what can only be described as the most anticipated set of manufactured conditions since copper’s Hamanaka affair of the mid-1990s. And those conditions will live on through this year at least.

As Indonesia’s government set new rules for its extractive industries to follow in late-September, prices of the element most at risk of not meeting those new rules by the target date of 31 December naturally surged. The lack of information – and the proliferation of misinformation – left the nation’s nickel mining and processing sector in the dark regarding what it could legally mine going into 2026. Betting on the worst outcome, global traders and price-setters considered an outright halt to nickel operations from the one country that offered up more than 60% of the world’s nickel. From a nine-month low of US\$14,118/t (US\$6.40/lb) on 16 December, LME cash rallied to US\$16,763/t (US\$7.60/lb) by 2 January 2026, before peaking at US\$18,738/t (US\$8.50/lb) at the end of that month.

That “deadline” for miners to lay out its extraction plans was extended to 31 March, but as we approach that date there is still a lot of confusion about how much and which type of ore miners can take out of the ground. Announcing a national total quota of 260-270 Mt of nickel ore for the year brought more questions than answers, and yet in short that is all we have to work with. As with last year there will be another opportunity for miners to re-submit their desired ore extraction volumes for approval in July. But unlike last year, the government’s “new process” is now an annual event, rather than once every three years. So, at some point in Q4, the uncertainties which underlay this new process from October last year will begin afresh once more.

For now, the market appears to have accepted the notional figure of 260 Mt for want of better information and has moved on. Any supply side constraint will make itself known later in the year, and this subject matter will come back into focus then.

The latest Middle East conflict has taken centre stage across a much wider range of commodities, with oil and derivatives, aluminium and copper being most exposed to events. At first consideration, nickel prices would not be directly supported by the conflict – indeed, by 24 March prices had fallen to their lowest levels since late-December 2025. But should the conflict drag on for longer than the “anticipated” two months, there will be consequences for both sulphur and certain catalysts supply which will begin to hold up Indonesian nickel production, especially the intermediates segment which often serves the downstream rechargeable battery manufacturing sector.

Given the way Indonesian nickel production has ramped up in recent years and is still expected to do so over the next two to three years, anything which holds up their growing contribution to the excesses seen in the market would be constructive from a price point of view. Indonesian intermediates are being upgraded to sulphate and/or converted to Class I metal with increasing ease, and now two refineries can ship to LME warehouses in the same way that the growing list of new Chinese cathode producers are just making metal for stock rather than for direct use. LME inventory now stands at over 280 kt compared with 160 kt in January 2025; equally, stock on the Shanghai Futures Exchange (SHFE) has almost doubled in that time, to 64 kt currently.

As prices declined during 2025 we were expecting to see some financial distress across several operations which could have permanently reduced global capacity. But it has not always gone that way. Chinese operator, Jiangsu Delong, fell on hard times which threatened the existence of Indonesian subsidiary, PT Gunbuster. But other shareholders took up Delong’s equity and Gunbuster survived. South32 confirmed that Cerro Matoso was on the block and, as with BHP’s Nickel West’s division, could have been put on care and maintenance if no buyer were found. But CoreX completed the acquisition of Cerro in December 2025, keeping its 30 ktpa capacity in the game.

Doniambo (60 ktpa in FeNi) in New Caledonia remains at risk, as does Madagascar’s Ambatovy (60 ktpa metal), but instead of financial stress taking Ambatovy offline this year, it was two typhoons in close succession that forced suspension. Its status remains under review. Equally, Sherritt’s Moa Bay (35 ktpa) and General Nickel’s Punta Gorda (20 ktpa) will both struggle this year given the energy crisis in Cuba since February.

Natural disasters and government intervention in Indonesia should keep supply flat this year at 3.9 Mt for the first time since 2016. We are assuming a “return to normal business” in 2027-2028 (+9% and +4%, respectively), but a lack of projects during the mid-term outlook from 2029 to 2032 keeps the lid on supply side growth at 2% a year on average. It is only from 2033 that this picks up (+3-4% a year) once again because of the need for new supply to be made available (our “additional adjustment”) and rising contributions from secondary batteries being recycled.

If, back in 2020, we had forecast an average annual global nickel consumption rate of almost 8% for the five-year period 2021 to 2025, we might have attracted some serious questions. Yet that is what we estimate has happened, and it is only because the growth in supply has exceeded that, at more than 9% a year, that the market has found itself holding on to considerable excess.

However, as with supply, demand has been extremely uneven, with China and Indonesia funding both sides of the coin in equal measure, leaving the rest of the world to fight over the scraps. For our mid-term outlook, we do not believe there is sufficient evidence that those patterns will change. For the most part, western markets for stainless steel and battery materials (the two critical markets for nickel) have remained weak; tariffs on trade, governments reversing progressive energy policies on the one hand and yet offering inadequate guidelines on renewables on the other will hamper battery materials progress for the next five years. Corporations are merely squeezing the assets that they have today to benefit share prices and dividends and will wait out investment for when it is deemed profitable.

In battery materials, sales of electric vehicles (EVs) are still rising but markets for nickel-containing materials are under pressure. Nickel's use in mid-level grades of precursor (pCAM) rather than trending up to the 811 and higher grades is growing as battery-consuming segments find applications more suited to cost and performance levels. But it is the rate of progress with lower-cost lithium-iron-phosphate (LFP) which continues to be nickel's principal threat. LFP cells are winning over those who once doubted their range and efficiency, despite the concern over the cost-effectiveness of recycling. As with stainless steel, China is dominant in the battery sector, but growth in demand for nickel-in-pCAM chemicals is slowing compared with the large double-digit percentages of 2022 and earlier. South Korean and Japanese players remain the “West's” main competitors, but they are finding it difficult to compete with China, while Europe's disjointed efforts at building a renewables sector will continue to delay the formation of partnerships which could have provided stronger growth.

After six consecutive years of calculated and anticipated surplus across global nickel markets during the period 2022-2027, there is hope for a more balanced position from 2028-2029. With an eye of where annual average prices have fallen to in previous periods of substantial oversupply – around the 70th percentile with spot lows potentially around the 50th percentile – we expect the underlying drift in nickel prices will be downward to US\$15,500/t (US\$7.00/lb) through to 2028 (2026\$). While this is not far below where prices sit now the longevity of such prices may yet be enough to force suspensions by producers that could enable the market to rebalance.

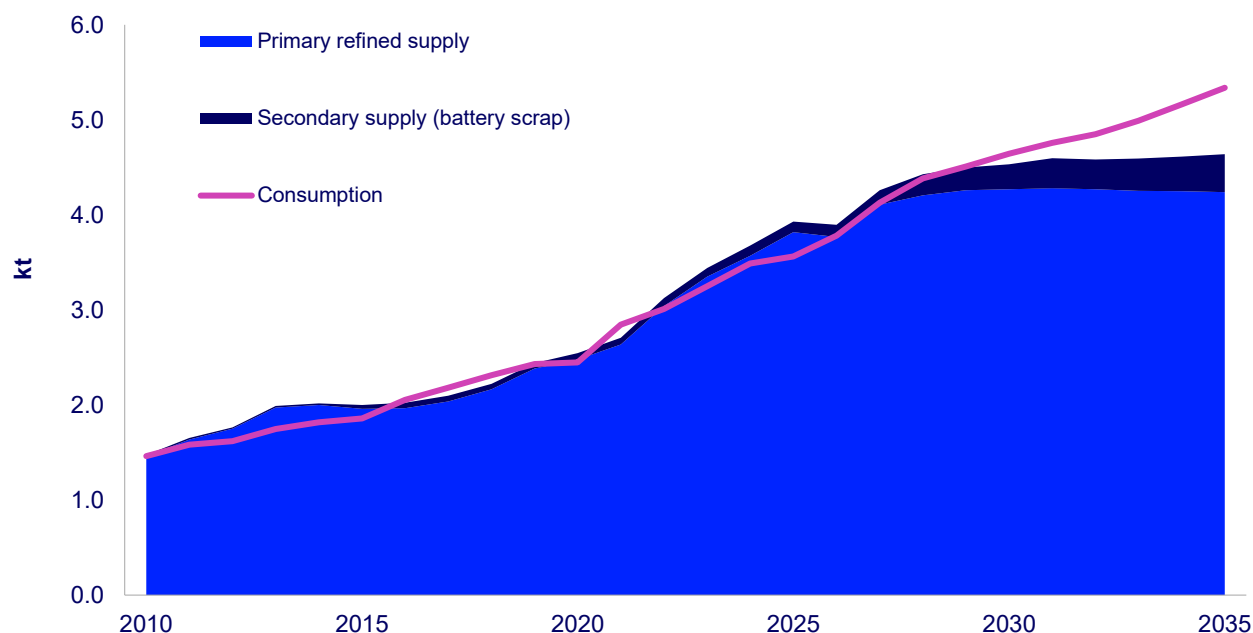
Given the anticipated supply side constraint thereafter, we expect prices to recover by 2030 when annual averages return to values above US\$16,900/t (US\$7.65/lb), with the subsequent period of undersupply pushing them onward to US\$24,000/t (US\$10.90/lb) by 2032. The market will need to start drawing from the pool of possible projects or as yet unidentified resources by then if supply is to keep pace with demand. By 2035, that requirement approaches 550 kt. The need for annual average nickel prices above US\$20,300/t (US\$9.20/lb) by then will help incentivise such projects but as the market rebalances, we anticipate later prices to settle around our long-term price of US\$19,600/t (US\$8.90/lb).

Price forecasts and refined market balances

	2025	2026	2027	2028	2029	2030	2031	2032	2033	2034	2035	2036
kt												
Mine prodn	4206	4111	4596	4822	4853	4872	4867	4952	5093	5248	5412	5574
Smelter prodn	3910	3810	4278	4514	4558	4571	4572	4667	4806	4961	5111	5256
Primary refined prodn	3929	3895	4259	4389	4443	4455	4467	4557	4688	4835	4977	5114
Secondary refined prodn	0	0	0	38	55	76	128	125	155	178	212	238
Total refined prodn	3929	3895	4259	4428	4498	4531	4595	4682	4842	5012	5189	5352
Consumption	3562	3782	4130	4385	4510	4644	4758	4849	4990	5161	5338	5479
Balance	367	113	130	42	-13	-112	-162	-167	-148	-149	-149	-127
Stocks (implied)	2161	2275	2404	2447	2434	2321	2159	1992	1844	1696	1547	1419
Days of consumption	221	220	213	204	197	182	166	150	135	120	106	95
LME price												
Current Usc/lb	687	780	723	731	775	867	961	1093	1184	1220	1220	1196
Real Usc/lb	709	780	708	702	730	800	870	970	1030	1040	1020	980
Current US\$/t	15154	17203	15947	16112	17096	19110	21197	24106	26109	26890	26901	26363
Real US\$/t	15639	17203	15619	15471	16094	17637	19180	21385	22708	22928	22487	21605

Source: Wood Mackenzie

Requirement for new refined nickel production



Source: Wood Mackenzie

Nickel sulphate prices and premiums

Over the medium to long term, we forecast that the China domestic nickel sulphate premium will gradually ease due to ample intermediate supply growth and comparatively weak demand growth from nickel-based pCAM (the primary end-use for nickel sulphate). We also forecast nickel MHP payability, the main feedstock for Chinese sulphate producers, will gradually decline from 2026 onward, falling to a low, stable level by 2030, driven by rapid intermediate supply expansion from Indonesia. Our long-term MHP payability forecast is for it to stabilise at around 86–87% by 2035.

Domestic demand for nickel-based pCAM is constrained by two key factors: the rising adoption and performance improvements of LFP batteries, and the subdued growth in nickel demand from downstream battery sectors in China. Specifically, China's growth rate in EV production has slowed, with nickel demand from power batteries nearly saturated and its growth decelerating. Compounding this, energy storage batteries, which are experiencing significant growth momentum in China, do not utilise nickel-containing batteries.

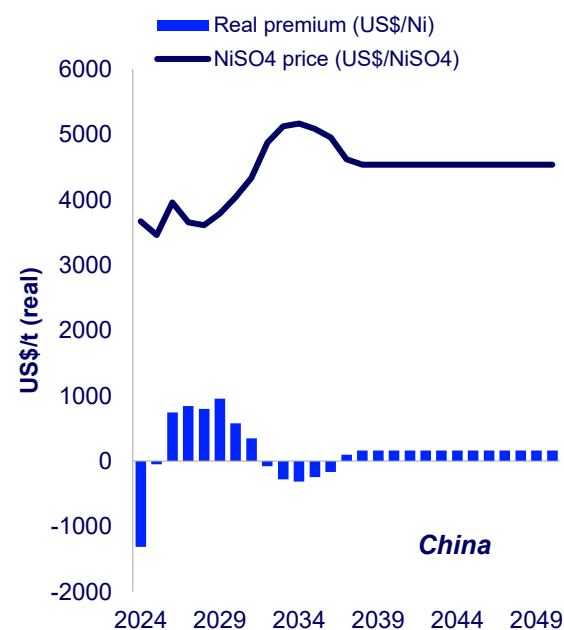
Together, these demand-side headwinds mean sulphate producers will face challenges in passing on higher prices to pCAM and battery manufacturers. Though we might see the production of solid-state batteries with high nickel content chemistries pick up in the next three years, which is expected to boost demand for high-nickel pCAM, there remains uncertainty around both large-scale production and the actual volume of nickel demand growth that such batteries will bring. As such, the upside for nickel sulphate prices is expected to fade after the near-term period, leading to a gradual and durable return to a discount versus LME nickel.

From 2032 onward, China's nickel sulphate premium will fall into negative territory, with the discount widening to stabilise at around US\$200–300/t nickel by 2035. This mid-term discount reflects the persistent imbalance between expanding intermediate supply and constrained demand, which will weigh on pricing. In real terms, the discount is projected to stabilise at around US\$250/t nickel by 2035.

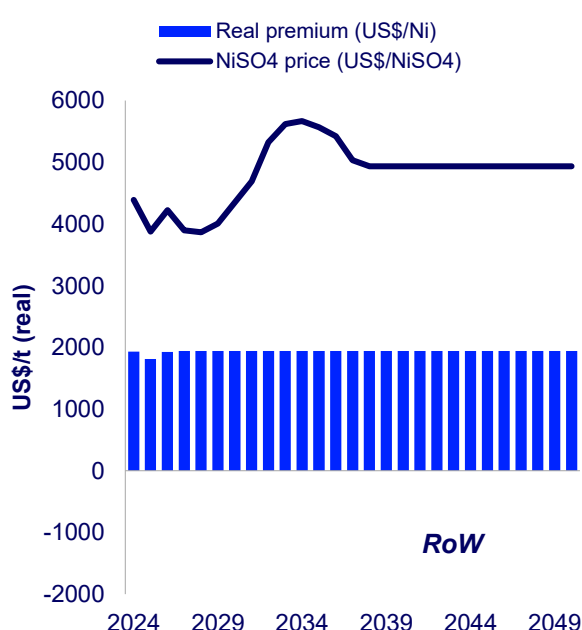
Outside China (and Indonesia), where metal dissolution remains a dominant production route, nickel sulphate commands a substantial and stable premium. The RoW premium is estimated at around US\$1,900/t nickel in 2026 and will hold steady at this elevated level through 2035. This stability is driven by the continued reliance on metal dissolution in these regions, which supports higher production costs and sustained premium pricing.

Over the long term, should regions outside China gradually shift away from metal dissolution toward intermediate-based production, we expect a similar pricing dynamic to that seen in China: at best moving toward parity with LME nickel, with a shift to a persistent discount becoming increasingly likely in the long run.

Nickel sulphate prices and premiums



Source: Wood Mackenzie



Source: Wood Mackenzie

Risks to forecast

Short-term upside

- War in the Middle East disrupts the supply of sulphur to Indonesia thereby depriving HPAL plants of a crucial raw material needed to generate energy and produce sulphuric acid for leaching. Indonesia depended on the

Middle East for 75% of its sulphur supply in 2025 and the consensus is that existing stocks may only last until the end of April.

- Closure of any of the world's major producers would provide price support. Questions remain over the future of Doniambo (New Caledonia), Ambatovy (Madagascar), Fort Saksatchewan (Canada) and Punta Gorda (Cuba).
- The momentum building in China's stainless industry after the Spring Festival holiday continues through Q2 with meaningful improvements in market demand and steadily increasing production rates.

Short-term downside

- Comments from Indonesia suggesting that RKAB quotas will be higher in 2026 than originally indicated (including anticipated upward revisions in quotas for some companies in July), enabling mine feed to smelters and HPAL plants to increase and, therefore, production from these plants to extend recent oversupply.
- Indonesian imports of nickel ore from Philippines increase above last year's level signalling that the restraint put on smelters and HPAL plants by limiting domestic mining via RKABs may not hold.
- A global energy crisis stemming from the war on Iran and closure of the Straits of Hormuz undermines economies and heightens risk aversion, thereby weakening markets across commodities. Stainless plants in particular will be affected by weaker demand and higher energy costs.

Medium-term upside

- Economies recover, supported by resolutions in Ukraine and the Middle East, so demand strengthens.
- Deceleration in nickel production and expansion across the Indonesian nickel segment – as production cuts and project deferrals – and clarity of those plans, to help bring the nickel market back to balance. This might be facilitated by further tightening on permits for mining (RKABs), for environmental reasons or to better control the exploitation of remaining reserves.
- The expansion of defence budgets across many countries in light of rising global aggression will boost nickel consumption in the non-ferrous segment of demand. Equally, commercial aerospace expansion and parts replenishment in the aftermarket will use more nickel.

Medium-term downside

- Increased use of stainless steel scrap in China above existing forecast, so that growth in nickel consumption in Chinese stainless – the main driver of global growth – falls behind growth in Chinese stainless production.
- Global trends in batteries for electrification continue to move away from high-nickel chemistries as LFP becomes more technologically capable and other technologies start to penetrate the market.

Long-term prices

The project incentive price (high case long-term price forecast)

A price that will generate the required return on capital for new operations is commonly referred to as the project incentive price and is often used as a surrogate for forecast long-term metal prices. This is premised on the broad assumption that new production required to meet market demand will not be forthcoming unless the market delivers a price that generates a satisfactory return on capital. Over the longer term, annual prices will therefore default to the long-run project incentive price.

As a theory it is intuitively sound but possibly ignores the real-world dynamic. New operations are commissioned on price expectation. However, once in production, pre-production capital costs are sunk and neither the capital cost nor the previously required return on capital are relevant to future production decisions for that operation. Once

commissioned, production levels for individual operations are driven by the need to sustain positive cash flow and profits.

This explains the poor historical correlation that exists between actual prices and incentive prices for the nickel industry. From 2002 to 2010 the market delivered a nickel price that was on average 37% higher than the project incentive price. From 2011 to 2020, the nickel price averaged 35% below the project incentive price.

While the past 10-year market price would imply that no projects would get developed, this has not been the case. Indeed, we have seen the largest growth in nickel supply the industry has ever seen over that period, growing by 54%, from 1.65 Mt to 2.54 Mt in 2020. This compares with just 27% in the prior ten-year period where, in theory, prices should have incentivised many projects to fruition.

However, during the second half of the 2001 to 2010 period we witnessed the advent of NPI production in China. This in itself was born out of adversity. The slow pace of development in the west of a number of large projects which were intended to supply the growing stainless steel sector in China, meant that China had to go it alone. And China has continued to do so since then. Firstly, in China, and now Indonesia, China has rapidly delivered new projects to the extent that forecast nickel-in-NPI production in Indonesia increased from 3 kt in 2013 to 1.4 Mt in 2023, and to 1.9 Mt including nickel in intermediates. It is this capability to deliver nickel units quickly, and at far lower capital costs than anything in the west, which has kept western new project development almost at a standstill. Nickel prices, while having risen significantly, have been volatile and not exhibited a certainty of remaining high for any period of time, leaving lenders unconvinced that a nickel project is going to be a good investment.

Consequently, under current market conditions, the project incentive price must be considered as the “high case” for our long-term price forecast.

Currently, our supply and demand analysis shows that by 2041 (taking a 15-year time horizon for nickel project development) we need an extra 1.45 Mtpa of new production capability, of which 815 kt is required by China. Assuming China continues to provide the nickel units it requires, regardless of current or future prices, this leaves 630 kt to be delivered by new projects in the rest of the world.

We calculate the incentive price at a 15% internal rate of return, all equity and pre-tax. The price to incentivise the RoW level of new project production is US\$36,375/t (US\$16.50/lb).

Our HIGH case, long-term, real-dollar price is therefore US\$36,375/t.

The price/C1 operating cost relationship – low case forecast

It is rare for supply to precisely match demand. When the market moves out of balance and generates unsustainable surpluses or deficits, decisions to decrease or increase production are determined by the profitability of producers – thus changes to the metal price should drive the system back to equilibrium. The nickel industry is slow to react, with producers preferring to “sit it out” unless there is a real perception that low prices are going to prevail for a lengthy period of time. The breakeven percentile of the annual nickel price on the nickel industry C1 cash operating cost curve is a good measure of this profitability.

For the two decades from 2001 to 2020 the average breakeven position was the 91st percentile i.e. 91% of production generated positive operating cash flow and 9% of production incurred negative cash flow. Over the 2011 to 2020 decade the average breakeven percentile was the 86th percentile. More recently, the breakeven percentile has been at or higher than the 95th percentile. We therefore use the average of 95th percentile of costs to represent the breakeven price. We consider that a breakeven price derived from future C1 cost curves (the M1 margin is equivalent to EBITDA), comprising a population of base case operations and probable and possible projects, to be a minimum sustaining price for the nickel industry.

Using this methodology, our LOW case, long-term, real-dollar price is US\$18,560/t (US\$8.40/lb).

The price/C2 operating cost relationship – base case scenario

We consider that a healthy mining sector must deliver not just positive C1 operating cash flows but profits. Satisfactory profits will ensure not only that production is sustained but will also ensure that new investment funds will flow to the industry. The measure for profitability is the C2 cost curve which is the C1 operating cost plus depreciation. We consider that a breakeven price derived from future C2 cost curves (the M2 margin, equivalent to EBIT), comprising a population of base case operations and probable and possible projects, to be the most appropriate measure for a

long-term price. Over the past three years, the nickel price has averaged at the 90th percentile of the C2 curve and we therefore use the 90th percentile for the C2 curve relationship.

Therefore, our BASE case, long-term, real-dollar price is US\$19,620/t (US\$8.90/lb).

Long term prices in US\$, real	Basis	Nickel price US\$/t Ni	Nickel price US\$/lb Ni
Base Case	C2 operating cost	19,620	8.90
Low case	C1 cash cost	18,560	8.40
High case	Project incentive for 0.75Mt/a	36,375	16.50

Source: Wood Mackenzie

Demand

Overview

Global economies continue to struggle in the face of ongoing political upheaval and uncertainty, which in the past month originates from the conflict between US/Israel and Iran. With an almost complete closure of the Strait of Hormuz, oil prices – and therefore energy prices – have sky rocketed, although nickel observers are perhaps more focused on the similar impact on sulphur shipments on which HPAL plants around the world, especially those in Indonesia, depend. Crucially, as well, sentiment has taken a dive just as markets were anticipating some improvement, and aversion to risk remains.

For nickel, global consumption increased by 2% in 2025, to 3.56 Mt, reflecting slower markets in China where stainless demand was weak and prices were low, while the growing preference for LFP batteries for electrification undermined the use of nickel in ternary chemistries. Elsewhere, stainless markets were poor with volumes still well below pre-pandemic norms.

Europe in particular had another bad year as stainless contracted for the third year out of four. 2026 should be better – geopolitics and economies allowing – as the advent of CBAM and the anticipation of more severe safeguards by mid-year should give a boost to stainless mill activities. But despite improvement, the future is one where regional output never returns to earlier highs. On the plus side, recent advances into special and nickel-base alloys have protected some of the mills financially recently, and this will likely continue as demand from aerospace, defence and military applications increases.

The US fared better last year, as tariffs reduced imports and allowed domestic stainless mills to increase output. However, while that tariff support remains there is limited room for volumes to rally again. In the US too, dependence on the nickel alloys sector is increasing.

Given the worsening geopolitics centred on the Middle East, our forecast of 6.2% growth in global nickel consumption in 2026, to 3.8 Mt, appears a little optimistic. That said, most of that growth is driven by China (more stainless and sustained growth in nickel ternary batteries), Indonesia (more stainless capacity coming, as well as battery materials) and India. Primary nickel uptake in stainless will increase by 6%, to 2.4 Mt, while that in non-stainless will rise by 7%, to 1.4 Mt, most of that in battery materials (up 10% to 600 kt).

After the turmoil of 2026, a rebound in demand in 2027 may strengthen growth in that year before slowing to see out the decade. Global nickel consumption by then will stand at 4.62 Mt, which is subsequently expected to increase with a CAGR of 2% over the long term, to 6.73 Mt in 2050. By then, the share of demand taken by stainless steel will account for less than 50% (compared with 63% in 2026) whereas that of battery materials will account for 36% (16%).

Regional nickel consumption

										CAGR	Change kt	CAGR
	2025	2026	2027	2028	2029	2030	2035	2040	2050	'25-2035	'25-2035	'25-2050
Europe	274	283	295	305	314	331	363	367	376	2.9%	89	1.3%
United States	113	120	123	126	129	133	145	156	190	2.5%	32	2.1%
Japan	176	176	179	182	184	187	196	201	221	1.1%	20	0.9%
China	2310	2431	2617	2774	2850	2926	3412	4017	4523	4.0%	1102	2.7%
Indonesia	405	449	560	614	622	630	671	700	757	5.2%	266	2.5%
South Korea	95	117	133	148	161	174	225	231	235	9.0%	130	3.7%
Others	189	206	224	237	250	263	325	360	426	5.5%	136	3.3%
Global Total	3562	3782	4130	4385	4510	4644	5338	6033	6729	4.1%	1775	2.6%

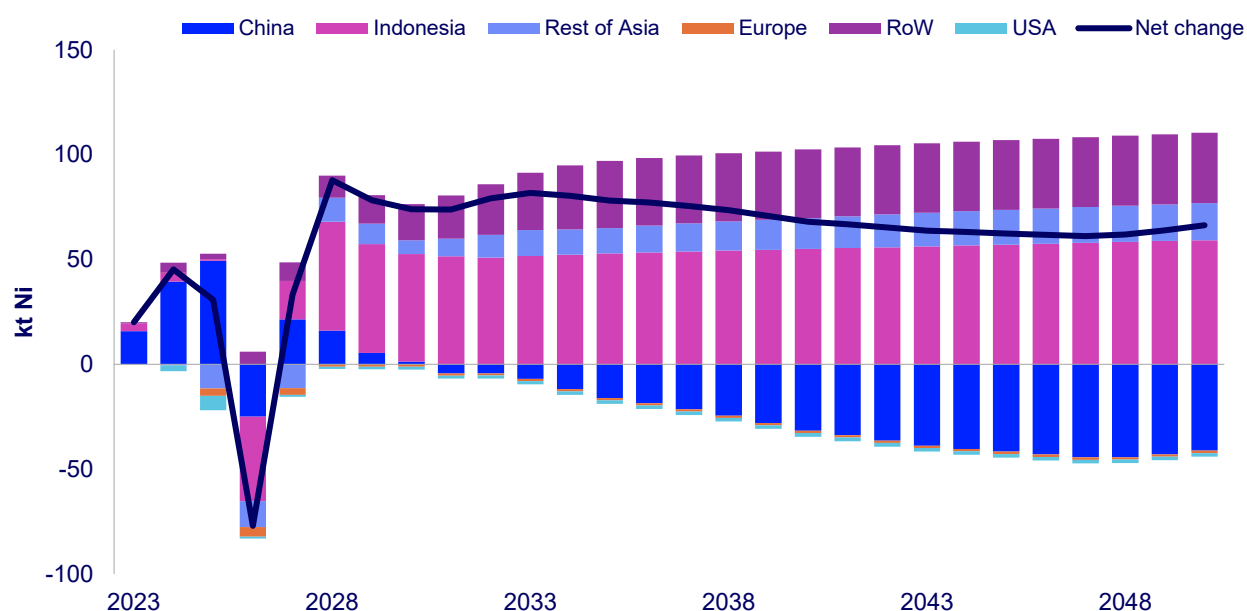
Source: Wood Mackenzie

Changes since the last quarter

The main changes since the previous quarter relate to nickel demand in stainless steel (globally) and battery chemicals (Asia and Africa):

- The increase in estimated Chinese data for 2025 is tied to stronger stainless while lower 2026 forecasts are mostly due to a downward revision in nickel in battery pCAM. Over the long term the net reduction in Chinese demand is due to a lowering of forecast nickel consumption in battery chemicals.
- A lower forecast for Indonesia in 2026 is due to Jiangsu Delong's Obsidian stainless mill being offline and uncertainty over its re-start. Further out the ramp up of new stainless capacity from Jindal (2026) and Eastern (Zhenshi; 2027) as well as the assumed return of Obsidian boost Indonesia's production to levels above prior forecasts.
- The net rise in long-term consumption in the Rest of Asia reflects the upward revision in Indian stainless production from the late 2020s.
- Forecast consumption for the rest of the world has increased over the mid- to long-term because of a positive correction to our near-term forecast of stainless production at Columbus in South Africa (through to 2028) and the ramp up of at least two new pCAM plants in Morocco.

Main changes in nickel consumption outlook since the previous quarter



Source: Wood Mackenzie

Nickel in stainless steel and other alloys

Africa

With only one stainless steel producer in Africa - Acerinox's Columbus mill in South Africa – the outlook for the region depends on the balance in production between stainless and carbon alloy steel. The latter was added to the product mix in 2020 due to the rise of Indonesian stainless and the decline of Columbus's shipments of hot-rolled coil to Bahr in Malaysia, which Acerinox sold in 2024. A year ago, Acerinox's plan was to focus more on southern African markets so that export volumes would shrink from around 70% of the plant's business to only 30%. At the same time, electrical steel was added to the product mix while stainless would be limited to ferritic grades only, thereby severing the plant's exposure to volatile nickel prices. Our forecasts are based on this proposed strategy leaving stainless output capped at around 300 ktpa with primary nickel consumption falling to zero from 2028.

Asia

Asia will remain the driver of mid- to long-term growth in global stainless as China, Indonesia and India are the only countries where we anticipate growth to be sustained into the mid-term and beyond.

In **India**, Jindal is planning to expand into more specialised stainless grades and alloys to take advantage of the

nation's infrastructure and energy transition roadmap that will need more value-add, high-spec products in rail, auto, defence and corrosion resistance applications. The government also wants to deepen its domestic manufacturing capabilities to reduce its dependence on imports. After increasing by only 1% in 2025, to 4.4 Mt, Indian stainless production is expected to increase with a CAGR of 5%, reaching 5.7 Mt in 2030 and 7.2 Mt in 2035.

The more mature markets of **Japan and South Korea** will remain steady at around 2.2-2.3 Mtpa each with the risk on the downside. **Taiwanese** melt production will also remain steady at around 700 ktpa. This is approximately half of typical production levels prior to 2018 because since then some Taiwanese mills have imported semis from Indonesia for cold rolling rather than melt and cast their own slab. This strategy is now well established and there is no reason to expect the situation to change.

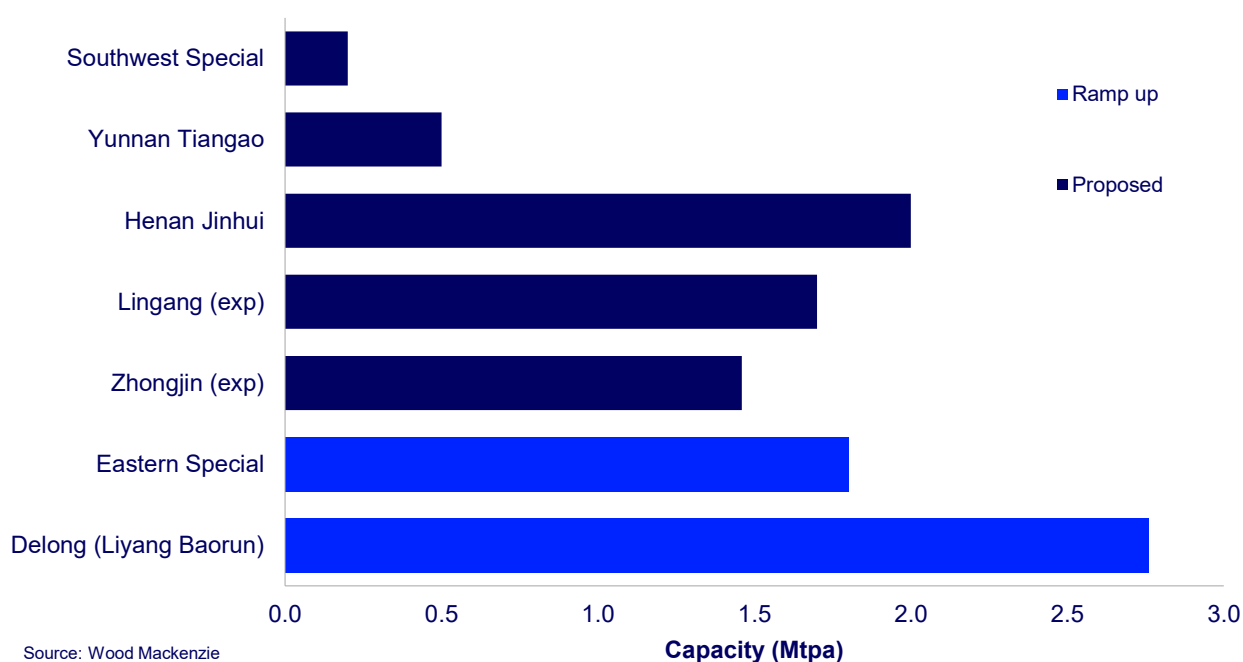
Regional stainless melt production increased by 2% in 2025, to 54.8 Mt, which will increase to 4% a year, to 68.6 Mt in 2030. We then expect growth to slow to 1-2% a year over the long term, reaching 85 Mt in 2050.

China

China will continue to drive increases in global stainless production. Output in 2025 of 40.6 Mt was around two thirds of the global figure, and this is expected to rise with a CAGR of 4% over the next five years, to 50.1 Mt in 2030. Of last year's total, Tsingshan accounted for around 40% and with Tisco accounting for another 20%, other companies have struggled to compete. Tsingshan has ramped up new capacity that the market did not really need and this ultimately kept prices low, resulting in its competitors cutting output.

One casualty is the project pipeline. With Jiangsu Delong still ramping up the 2.7 Mtpa Liyang Baorun meltshop, only Eastern Special Steel (Zhenshi) will add new capacity anytime soon when its new 1.8 Mtpa meltshop comes online later this year. Other expansions and new projects totalling 5-6 Mtpa look increasingly unlikely, particularly those of Henan Jinhui, Yunnan Tiangao and Southwest Special (Gangcheng), all of whom have produced very little in recent years as Tsingshan became ever more dominant.

The pipeline for Chinese stainless melt shop capacity



China's governmental strategy of "anti-involution" seeks to prevent exactly what has happened in stainless – filling excess capacity and pushing prices down – and replace it with a focus on value rather than volume. China's stainless industry is heading toward consolidation – the latest loss being Posco's divestment of ZPSS (to Tsingshan). So far, the effort by the mills to claw back lost market share is to expand their product mix to include more specialty grades of stainless (containing more nickel, such as 316) and specialty products, such as thick or ultra-thin plate. In other words, the move along to value-add products is already happening; and this will continue.

An alternative strategy may involve off-shoring. While Eastern starts up its new melt shop in China, it is also dismantling the old one to move it to Indonesia, where the company has invested in NPI. That integration will lower

cost substantially. Similarly, Yongjin, one of China's main stainless cold rollers has plans for an integrated stainless complex in Vietnam (see below).

With the home market being at an uncharacteristically low ebb, the export market was important to stainless producers in 2025. Low-cost Chinese stainless is competitive in international markets and over the past 10 years or more protectionism against Chinese stainless has spread. But despite tariffs and duties Chinese product is still taking share, especially in markets like Europe. But it is not getting any easier in the current climate of trade tariffs triggered by the actions of the Trump administration in the US. Europe's initiation of CBAM charges from January 2026 and the anticipation of more severe safeguards by mid-year has stymied the flow of Asian stainless (much of which is Chinese-origin) to Europe so far this year. But for how long? China has become adept at circumventing duties and will probably do so again as this battle continues.

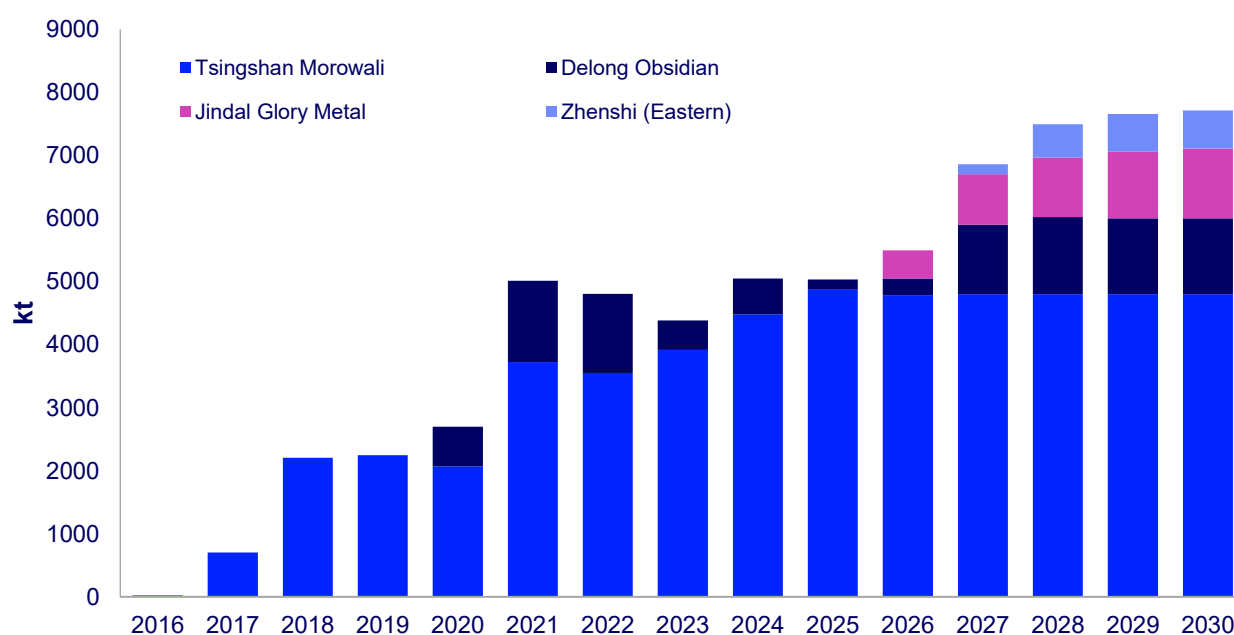
The main issue for nickel consumption in stainless going forward is how scrap use develops in China. With widespread access to such large volumes of NPI, both in China and Indonesia, the national average scrap ratio, at around 20%, is well below the 60-70% or more of "western" plants. But as domestic scrap becomes ever more available as the "old scrap" pile gets larger, and NPI supply becomes tighter, not least as Indonesia places greater preference on downstream products, such as stainless steel and nickel matte for the battery stream, scrap use will rise. The benefits of stainless steel scrap in terms of energy efficiency, recyclability and low carbon credentials will only enhance this trend. And as scrap use increases so primary nickel consumption will not keep pace with the growth in stainless production.

As China moves down the value-add path in stainless so it is also evolving its expertise in other alloys, especially nickel-base alloys. With Baosteel more than doubling its capabilities in this segment in 2024 we estimate that nickel consumption in non-ferrous alloys in China has increased from less than 100 kt in 2022 to 136 kt in 2025, and we see this increasing to around 160 kt by 2030. We believe this production is mostly intended for the domestic market and in any case certification – essential for aeroengines – will delay the penetration of Chinese product in international markets, but not forever.

Indonesia

Indonesia's stainless melting industry began in 2017 with Chinese investment from Tsingshan and continued in 2020 with Jiangsu Delong Obsidian. At the time there was a promise that further investment would increase both melting capacity and the number of operators but this has not taken place, perhaps because of the wave of protectionism outside China against low-priced Chinese stainless. Furthermore, while Tsingshan has expanded, Delong ran into financial difficulties and has produced only small quantities since 2023; Obsidian is currently offline.

Indonesian near-term consumption of primary nickel in stainless, by company



Source: Wood Mackenzie

However, new plants are coming to Indonesia. Jindal's new Glory Metal stainless plant (1.2 Mtpy) is due to enter commercial production any day now and may take advantage of Jindal's NPI joint venture (New Yaking) at Weda Bay. Also, as noted above, Eastern Special intends to move its old stainless mill in China to Indonesia where it will be fed by NPI from Zhenshi's two RKEF assets, Huabao and Yashi, the latter also in Weda Bay.

Finally, the Indonesian stainless industry is expanding and with competition in China this is something that may gather momentum in the years ahead. Assuming that Delong Obsidian does return to production and succeeds in recovering to more normal activity levels, the added new capacity will take national output from 5 Mt in 2025 to 7 Mt in 2030, with still more capacity available at Obsidian. Primary nickel consumption will increase by 50% over the same period, from 400 kt to 600 kt.

Vietnam

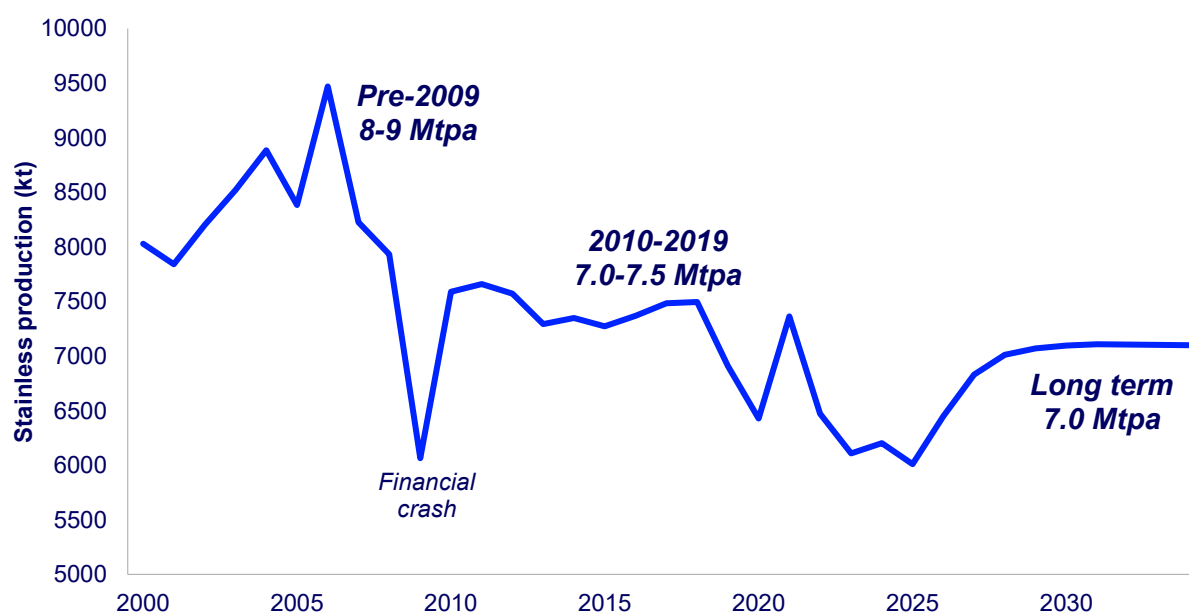
Vietnam may join Indonesia as a new centre of stainless steel melting expansion. There are no stainless melt shops currently in the country but with Yongjin Technology recently announcing plans to integrate its existing cold-rolling plant with hot-rolling and melting, this is going to change. Also on 9 February 2026, the Vietnamese government approved the Steel Industry Development Strategy which seeks to increase domestic production and prioritise higher value-add and environmentally sound products, including stainless steel and other specialties. The target is to produce 26 Mtpa of steel by 2030, and Yongjin's project is well timed to benefit from it; other projects may now follow.

Europe

After contracting in three years of the past four, European stainless melt production stood at 6 Mt in 2025, compared with 7.5 Mt in 2021, the last peak. It has become a cliché to say that the region's market will "bounce back this year" because this has not happened in recent years despite forecast optimism. However, with CBAM now physically charging importers and the EU expected to ratify more severe safeguards in July, imports from Asia, the main threat to domestic producers, have dried up. So, even though end user demand remains circumspect in the current geopolitical climate, the mills are starting to benefit from increased orders and slightly higher prices. For how long, remains open to debate.

Even so, our forecast 7.3% growth in 2026 may still prove optimistic, but this only takes production to 6.5 Mt, which still falls a long way short of traditional pre-pandemic levels of 7.0-7.5 Mtpa. In fact, whatever recovery is achieved, we expect mid- to long-term regional production to settle at around 7 Mtpa, with primary nickel use capped at 130 ktpa.

The outlook for European stainless steel melt production (excl. Russia and Turkey)



Source: Wood Mackenzie

There is no new melting capacity planned for "mainland" Europe, although de-bottlenecking and upgrading may facilitate some increases in existing capacity. However, new capacity is potentially coming in Russia and Turkey,

although in both cases initial investments are in stainless rolling with only longer-term plans involving vertical integration with melting.

In Russia, RNK (Russian Stainless Company, a joint venture between Chelyabinsk Electrometallurgical Plant, TMK, a Russian pipemaker and Nornickel), is building stainless cold-rolling and hot-rolling units that should become operational this year, with a melt shop (est. 600-700 ktpa) to follow in 2027. In Turkey, Saritas Celik is adding 800 ktpa of cold rolling to its existing stainless processing capabilities, due to start in 2027. Hot rolling (also 800 ktpa) could be added in 2028 with a similarly sized meltshop potentially installed by 2030. Therefore, it is possible that European stainless melt production will get boosted to 8 Mtpa or so in the 2030s.

While the challenge of imports has been seen off in early 2026 by CBAM, it is unlikely to go away for long. The impact of this competition and the weak markets of recent years have forced the region's main producers to shift their focus progressively toward value-add specialty products, including those with low carbon emission credentials specifically tailored for the energy transition. Most significantly, however, is the acquisition of stainless and nickel alloy producers by Acerinox (VDM in Germany and Haynes in the US) and Aperam (Universal Stainless and Alloy in the US), while Outokumpu has added nickel alloy powders to its product suite and intends to expand into special alloy production at Avesta in Sweden.

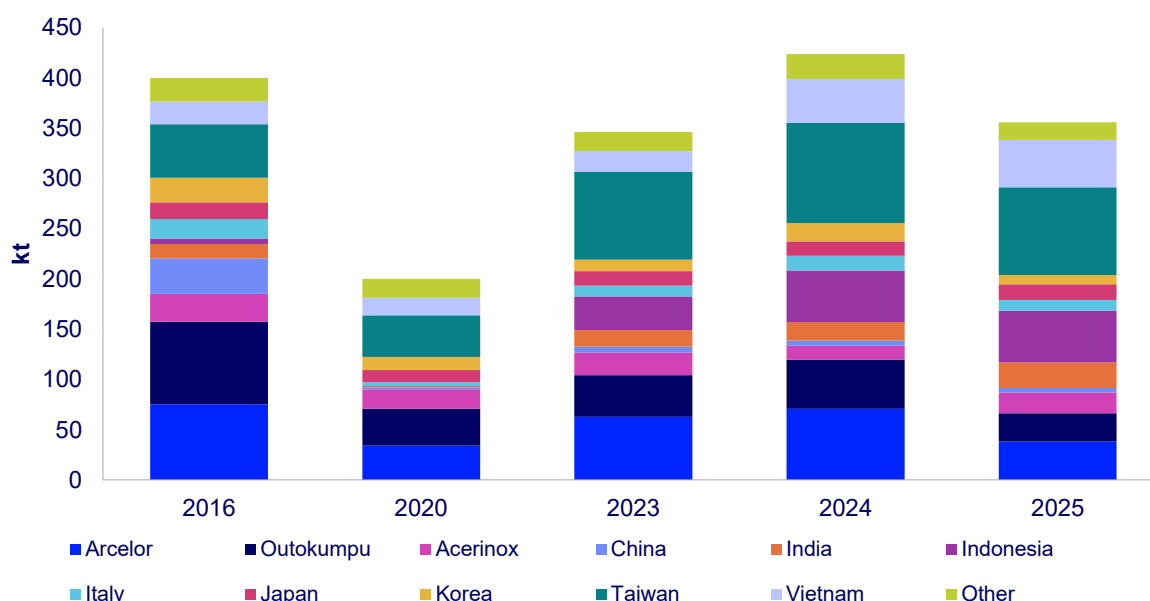
These moves have coincided with improvement in special alloy demand helped by the resurgence of the aerospace industry after Covid and currently the geopolitically induced interest in defence and military investment. And while stainless markets have remained weak, the special alloys business has been good for balance sheets. Assuming that imports of Chinese and other Asian stainless do not stay away for long, it is easy to see the Europeans concentrating more on specialties as they struggle to compete with imported commodity grades, although we are not quite there yet.

At least for now there is little competition from China in the specialty grades and nickel alloys. But again, with China expanding in these segments it might not be too long before the Europeans will need to work harder at protecting their shares of these niche markets.

North America

Stainless melting in the USA improved for a second consecutive year in 2025, this time by 7.6%, to 2.1 Mt, from 1.95 Mt in 2024, but it remains far off the recent high of 2.81 Mt set in 2018. In the first year of the new administration, imports of flat products (FP; hot/cold rolled >600 mm wide) fell by 16% to 356 kt while exports were little changed at 370 kt (-1.6%); long product (LP) imports fell by 4.1% to 165 kt while exports of LPs were down by 19.5% to 33 kt.

US imports of flat stainless steels (HS 7219)



Source: Wood Mackenzie, Global Trade Tracker

When placed against the rise in melt output, both sets of FP and LP trade imply greater home use of melted stainless, compared to 2024 when there were short-term contracts in place for higher-value steels for the export market.

However, trade data also suggest nickel imported for stainless melting fared no better than in 2024. Imports of Class II FeNi fell to 44.4 kt (gross) from 50.1 kt in 2024, with Brazil and New Caledonia providing the lion's share (65.5% and 24.5%, respectively), while *total* imports of Class I metal remained static at 80 kt. With little change also noted in non-stainless sectors, the data suggest that more nickel units for stainless melting were drawn from higher scrap use.

Stainless FP imports have risen or remained strong from countries which can produce basic, lower value stainless steels at lower cost than American domestics, confirming our belief from 2023 that US manufacturers have downgraded their exposure to low margin business. FP imports from India, Indonesia, Taiwan and Vietnam especially have displaced material from China (no doubt due to tariff and politically-acceptable sourcing issues), while imports from company groups with production assets within North America (North American Stainless, Outokumpu, Aperam) have fallen by up to half as their local sites increase utilisation rates.

Looking forward, we have kept stainless output largely in line with forecast industrial production, implying that if national production does exceed 2 Mt in any given year it will be because of short-term extenuating circumstances, rather than underlying gain from structural change. This suggests that nickel-in-stainless demand will remain stable at 25 ktpa over the long term.

Compared with stainless steel, a much greater amount of primary nickel is consumed in the American non-ferrous alloys sector, which includes superalloys. Currently this sector consumes 60-65 kt primary nickel, and we expect this will grow to 100 ktpa by 2040. But for this to happen it will need new investment in melting capability. Near-term problems run counter to that need. The commercial aerospace market has been beset with labour and management disputes for much of the past five years, and wider issues within US airspace (from changes to customer protections allowed under the current administration to cuts in staffing in airport control towers and TSA officials) do not encourage increased passenger use. But long-term, commercial aircraft fleets continue to expand and have an ongoing need for replacement parts. GE Aerospace acknowledged this last year by investing almost US\$1 billion at factories across 16 states, with half of that supporting the production and assembly of the narrowbody CFM LEAP engine. In mid-2025 start-up JetZero selected North Carolina as its location to build 20 Z4 jet airplanes per month by the late-2030s, with the first full-scale prototype designed to carry up to 250 passengers expected in 2027. And in late-2025 Williams International broke ground on a new turbine engine facility at Okaloosa County, FL, which should be commissioned in early 2027.

Finally, with hostilities seeing no end in Ukraine and now re-starting in the Middle East, we are already seeing increases in military budgets across North America, Europe and Asia. The proposed budget in the USA for 2027 of US\$1.5 trillion would represent a 50% increase on 2025-2026's allocation, with 25% increases expected for other countries from Russia to China, Germany and the Nordic countries.

Nickel in battery precursors (pCAMs)

Prior to 2020, the lithium-ion battery industry's trajectory had appeared set: battery makers were transitioning towards nickel-rich battery cathode chemistries (primarily NMC 811), driven by its superior energy-density performance – translating into longer-range EVs – and volatile cobalt prices, which incentivised battery makers to thrift out cobalt in favour of lower-cost nickel.

However, the launch of BYD's blade battery in 2020, underpinned by a next-generation version of lithium iron phosphate (LFP) technology, marked a turning point in this trend. Technological advancements enabled LFP to match the performance of mid-nickel cells for the first time, while delivering substantially lower costs and superior safety characteristics. Moreover, the relative price stability of iron-based inputs, compared with nickel's inherent volatility, further enhanced the economic case for LFP and LMFP adoption.

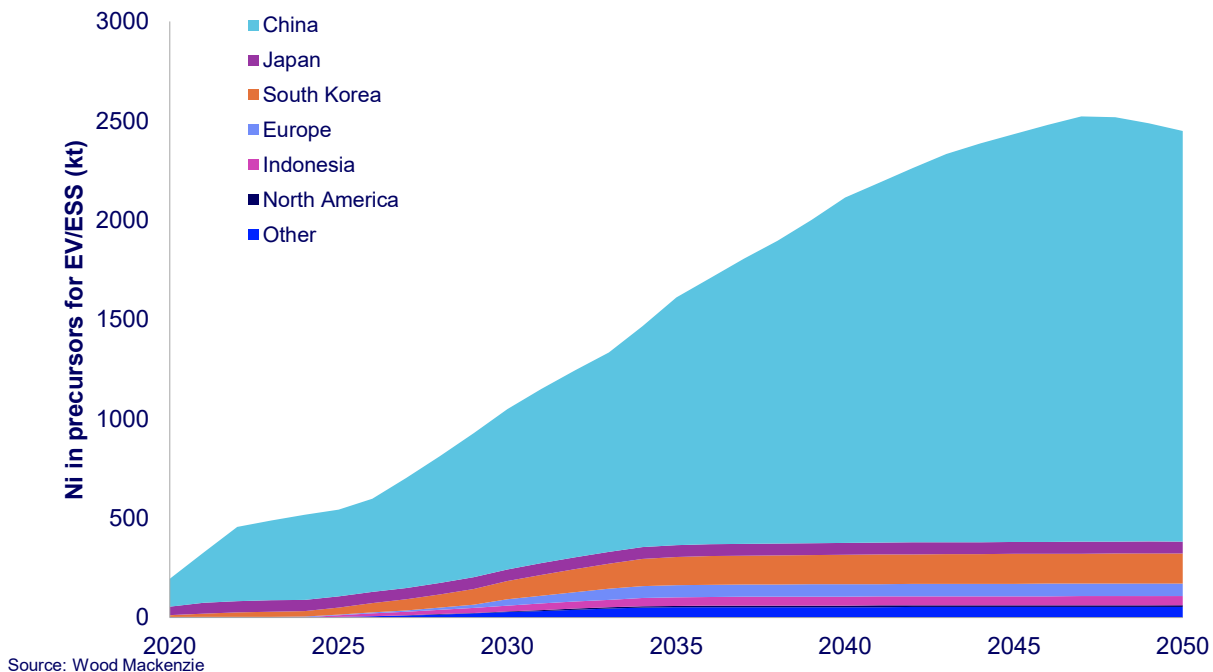
On a battery GWh basis, the global share of L(M)FP has increased from only 20% in 2020 to nearly 60% in 2025, largely at the expense of low-Ni and mid-Ni battery types. Aside from EVs, LFP has also benefited from the rapid deployment of battery energy stationary storage capacity worldwide in recent years, where its superior cycle life makes it the chemistry of choice for this segment.

LFP is expected to retain that share through to 2030 before retreating to around 50% of the market as the demand for high-Ni batteries increases to around 30%, while the penetration of newer technologies, such as Na-ion and solid state, also improves. Recent preferences for a high voltage mid-nickel battery has yet to gain significant momentum but could further undermine the growth in high-Ni (and potentially overall nickel demand from batteries).

China, naturally, continues to lead the way in the batteries-for-electrification segment both in LFP and nickel ternaries, and it remains dominant in the production of nickel battery raw materials: MHP, nickel sulphate, pCAM and CAM. But

the Chinese electric vehicle and battery segments face overcapacity and low utilisation rates, something the central government's recent "anti-involution" strategy – a focus on value rather than volume – seeks to rectify. By raising quality standards, the costs of compliance will increase and smaller, private players producing lower quality products will be forced out of the market; this is expected to result in market consolidation. SOEs on the other hand are already focusing on new technologies – eg, solid state – rather than expanding capacity.

Global nickel-in-pCAM consumption by country.



LFP makers are in a similar position. Such has been the speed of expansion as more players tried to take advantage of the surge in demand, that the market is now one of oversupply and low utilisation rates. As the government starts to back innovation to advance quality, the makers of cheaper, lower quality products will fall by the wayside.

In contrast, the ternary pCAMs and CAMs sector has so far naturally advanced along the value add path: low-Ni and mid-Ni have steadily lost out to high- and ultra-high-Ni (aided by the rise in LFP); looking ahead it is the high- and ultra-high-Ni types that the major players are prioritising for capacity build out. Indeed, the entire material supply chain – much of which is now based in Indonesia - seems to have evolved into one focused on value add: NPI (for stainless steel) is now routinely converted to matte for the battery stream; and while the expansion in MHP capacity for nickel sulphate production was originally all to feed the battery stream, excess sulphate is now converted to the highest value nickel product – metal - which in most cases goes to the LME and SHFE exchanges.

Aside from China, and Chinese investments in Indonesia, South Korean producers are making the biggest in-roads to international markets. Collectively Korean producers are second only to China and together with the Chinese are investing in raw materials and battery facilities in Europe and the Americas. But European and US backed projects continue to struggle with the balance between technical challenges and costs that have led to cancellation and bankruptcy.

While the European Commission and US government have put policies in place to incentivise investment in critical minerals, including nickel, there is a lack of focus across the entire supply chain, despite the current preference not to be leveraged to Chinese supply of one form or another. Hence, a mine project may get governmental support but only so far as the production of a concentrate or other intermediate that would then end up going to Asia for processing to sulphate or pCAM. The latest casualty in this regard is Vale's cancellation of the Becancour nickel sulphate plant because Posco and General Motors cancelled the pCAM plant which Becancour's sulphate would have served. The planned battery plant, due in 2026, remains but will be dependent on taking imported CAMs.

Therefore, of a global total of 543 kt nickel in battery pCAMs in 2025, China accounted for 80% and we do not see that proportion changing very much over the mid- to long-term. Global growth is slowing however, from a CAGR of 23% between 2020 and 2025, to 14% from 2025 to 2030, reaching 1.05 Mt by then, albeit with risks on the downside. By comparison, South Korea consumed around 36 kt in 2025 increasing to 92 kt in 2030. In contrast, Japan was a bigger

consumer in this segment, at 56 kt in 2025, but the lack of developments in that country means that figure barely changes by 2030. Europe and the US each consumed less than 5 kt in 2025.

Supply

Overview

Forecast refined nickel production will increase by 3.4% a year through to 2030, reaching 4.5 Mt (before adjustments) by that time. From there refined output is unchanged through to 2040 and then falls by 0.4% a year through to 2050. Refined nickel production from primary sources falls to 4.3 Mt in 2050.

Mined nickel production will reach 5 Mt by 2030, representing annual growth of 3% a year. Mine production then declines by 0.4% a year through to 2040, to leave mine output at 4.8 Mt by that time. Mine production declines to 4.4 Mt by 2050.

Regional nickel mine production capability

	2025	2026	2027	2028	2029	2030	2035	2040	2050	CAGR '25-2035	Prod'n kt '25-2035	CAGR '25-2050
Indonesia	2634	2640	3139	3356	3405	3419	3420	3420	3420	2.6%	786	1.0%
Philippines	494	364	322	317	308	308	308	308	278	-4.6%	-186	-2.3%
Russian Fedn	191	201	214	216	217	217	217	217	217	1.3%	26	0.5%
Canada	131	153	161	167	152	146	136	52	0	0.4%	5	-100.0%
Australia	55	49	40	40	40	40	40	40	0	-3.2%	-15	-100.0%
N. Caledonia	143	192	206	208	208	208	208	208	208	3.8%	65	1.5%
Others	558	580	604	610	616	626	536	530	290	-0.4%	-21	-2.6%
Global Total	4206	4180	4686	4915	4947	4965	4866	4775	4413	1.5%	660	-2.6%

Source: Wood Mackenzie

Regional nickel refinery production capability

	2025	2026	2027	2028	2029	2030	2035	2040	2050	CAGR '25-2035	Prod'n kt '25-2035	CAGR '25-2050
China	1202	1229	1250	1285	1274	1271	1271	1271	1271	0.6%	69	0.2%
Japan	112	107	107	107	107	107	107	107	107	-0.4%	-4	-0.2%
Indonesia	1846	1846	2139	2203	2253	2256	2257	2257	2227	2.0%	411	0.8%
Norway	102	100	100	100	92	92	92	92	92	-1.0%	-10	-0.4%
Russian Fedn	123	123	132	137	142	142	142	142	142	1.5%	19	0.6%
Canada	125	128	134	134	134	134	133	90	60	0.6%	8	-2.9%
Australia	35	34	35	37	37	37	37	37	0	0.7%	3	-100.0%
Others	385	395	449	476	494	507	477	477	395	2.2%	92	0.1%
Global Total	3929	3961	4346	4479	4534	4546	4517	4474	4295	1.4%	588	0.4%

Source: Wood Mackenzie

Changes since the last quarter

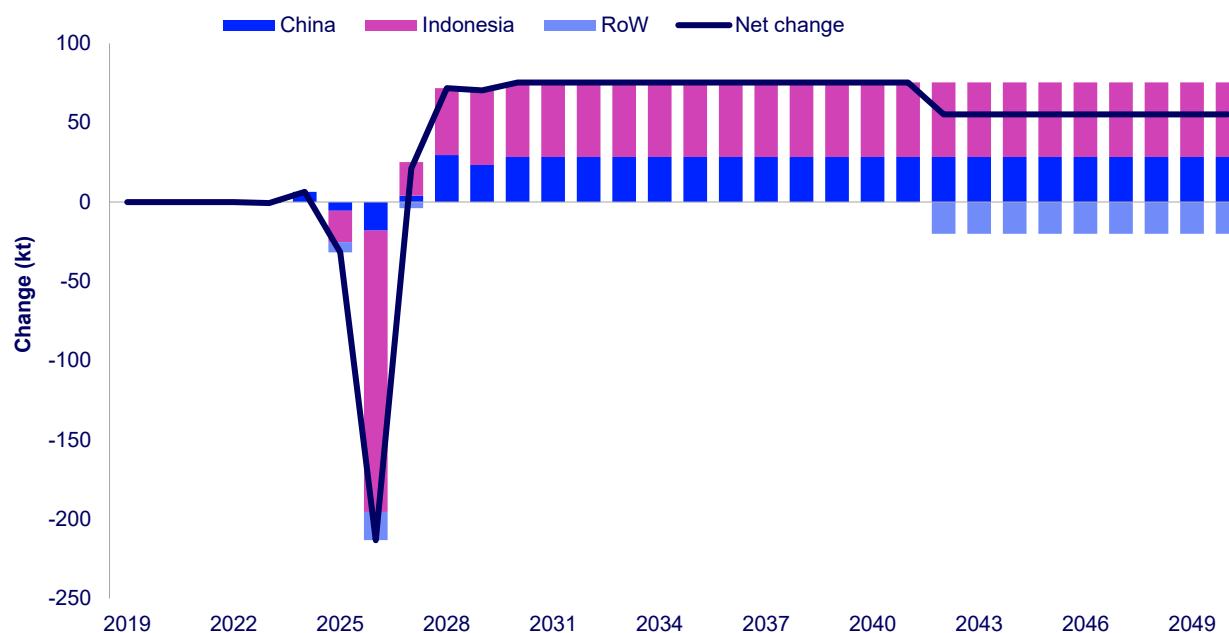
Supply changes since Q4 2025 mainly reflect expectations of lower NPI output in Indonesia, following confirmation in February that Indonesian RKAB mining quotas will be limited to 260-270 Mwmt.

The largest cut in output of 130 kt in 2026 has been attributed to operations at the Weda Bay Industrial Park (IWIP) linked to the Weda Bay Nickel (WBN) mine - operated in a joint venture between Tsingshan and French miner Eramet - after it was confirmed in February that the mine's initial RKAB quota would amount to just 30% of what it had requested. IWIP's output has been downgraded over the forecast period, reflecting a tighter regulatory environment in Indonesia.

The inclusion of Nickel Industries' Excelsior refinery in Indonesia and the Caoheidian expansion in China has boosted finished nickel output by 29 kt in 2026, and up to 87 ktpa over the forecast period.

Outside China-Indonesia, the Tagaung Taung FeNi smelter in Myanmar (20 ktpa) has been removed from the base case after export data appears to indicate that the operation has closed. Meanwhile, Pronico's FeNi smelter in Guatemala has been reinstated after the company outlined plans to re-start production in April 2026.

Main changes in refined nickel production since the previous quarter



Source: Wood Mackenzie

The supply landscape

The supply-side of the nickel market has been structurally transformed in the past twenty years. Nickel supply is now highly concentrated in a single country of origin, Indonesia, which accounted for 63% of global mine supply in 2025 (up from 7% in 2015), akin to levels seen in minor metal markets, such as cobalt. Indonesia's share of global mine supply will continue to steadily increase throughout the forecast period, climbing to 70% in 2035 and 78% by 2050.

At the refined level, Indonesia's position is less all-encompassing but still substantial, accounting for 47% of global output in 2025, and increasing to ca. 50% through to 2050. China and Indonesia together are forecast to account for 78% of global refined output in 2035, rising to >80% by 2050.

The structural changes relate not only to where nickel is produced geographically, but also the type of final nickel product being made. Exchange-deliverable Class I (>99.8% Ni) nickel supply, which historically (up until 2018) captured the lion's share of global refined output, now accounts for a minority share of the supply landscape (44% in 2025), owing to the rapid growth of non-exchange deliverable Class II (primarily NPI) output, first in China and later Indonesia, since the late 2000s. However, the advent of product optionality at operations in China and Indonesia (which includes capacity to convert Class II NPI into a matte for use in chemical/metal refining) has eroded some of the traditional distinctions between these two classes of nickel, and changes to the refined product mix are now likely to be reactive to demand-side fundamentals through the forecast period.

In addition, the rate at which production capacity has been added, particularly in the past ten years (mine supply averaged annual growth of 7% between 2015-2025), has been unprecedented. The pace of additions is expected to slow somewhat through to 2035, averaging annual growth of 3.5%.

Indonesia's nickel boom: 2014-to-now

Indonesia closed its doors to exports of nickel ore - first in 2014, and later 2020 - to catalyse investment in value-added nickel processing, aiming to maximise potential revenues from its mineral resources and spur domestic industrial development (firstly refined nickel processing through to stainless steel production, then later targeting a battery supply chain ecosystem). The result has been a spectacular transformation of both the nickel market and the country's mining sector, as (primarily Chinese) investment has poured into the Southeast Asian country's burgeoning nickel industry.

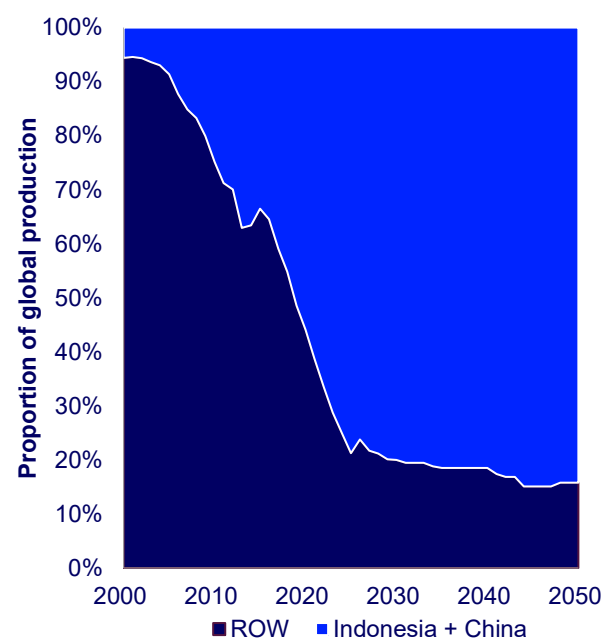
The initial phase of investment (post-2014) targeted NPI production capacity to feed China's domestic stainless-steel industry and a swathe of new Chinese-owned mills located within sprawling industrial parks (primarily the Tsingshan-backed Morowali Industrial Park (IMIP) and Weda Bay Industrial Park (IWIP)), benefitting from captive power supply

and port facilities, across the Indonesian archipelago. These developments saw Indonesia's refined nickel output grow from 102 kt in 2014 to >1 Mt for the first time in 2021. As this capacity has matured (Tsingshan alone operates ca. 500 kt of NPI capacity in Indonesia as of 2025) China's domestic NPI players have seen their output steadily decline since 2019, facing margin pressure from Indonesia.

The second investment wave (post-2018) shifted focus towards high-pressure acid leach (HPAL) plants to feed growing demand from the lithium-ion battery industry. Chinese firms have demonstrated their engineering prowess, by constructing multiple HPAL operations at a scale (often >100 ktpa Ni capacity), ramp-up timescale (often taking 1-2 years to exceed nameplate capacity), and capex intensity (around US\$30,000-35,000/t Ni vs US\$100,000/t Ni in the west) previously considered unimaginable by leading miners in the west. The first of these next-generation HPAL operations came online in 2020 and has been followed by a slew of similar such investments from Chinese firms and partners, taking Indonesia's total HPAL capacity from zero in 2019 to >500 kt nickel by 2025.

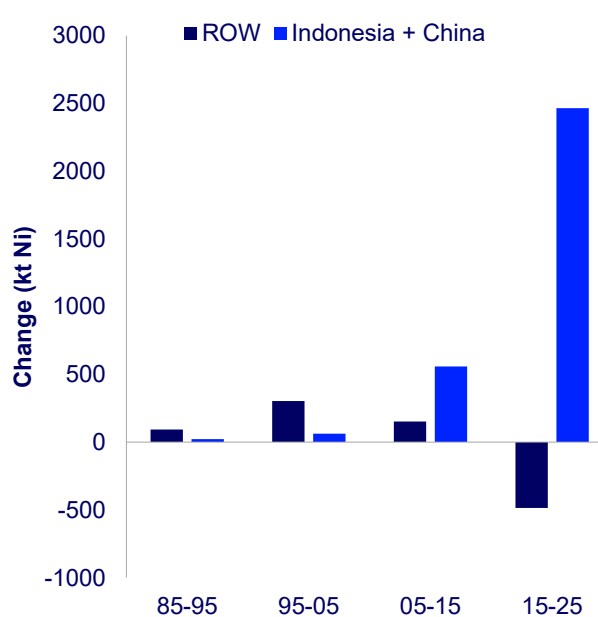
An important development since 2022-23 has been the evolution towards product optionality. Sino-Indonesian operations are now able to produce exchange-deliverable Class I nickel metal (cathode), via MHP from HPAL and matte via NPI conversion, thereby bridging the gap between Indonesia's vast nickel industry and internationally traded markets (LME/SHFE) for the first time, thus bequeathing the Sino-Indonesian nickel complex with greater influence over global price benchmarks. Indonesia will add a further 550 ktpa of nickel capacity from HPAL across 2026-27, extending the country's position in the Class I segment through 2030 and beyond; Sino-Indonesian Class I supply will overtake supply from the rest of the world by the mid-2030s.

Global nickel production



Source: Wood Mackenzie

Production change by decade



Source: Wood Mackenzie

RKAB quotas add layer of supply complexity

With Indonesia's supply-side dominance now firmly assured, the government has begun to utilise a burgeoning armory of policy levers to exert greater control over the country's resources, extend its oversight over the mining sector, and manage global supply and demand (in)balances.

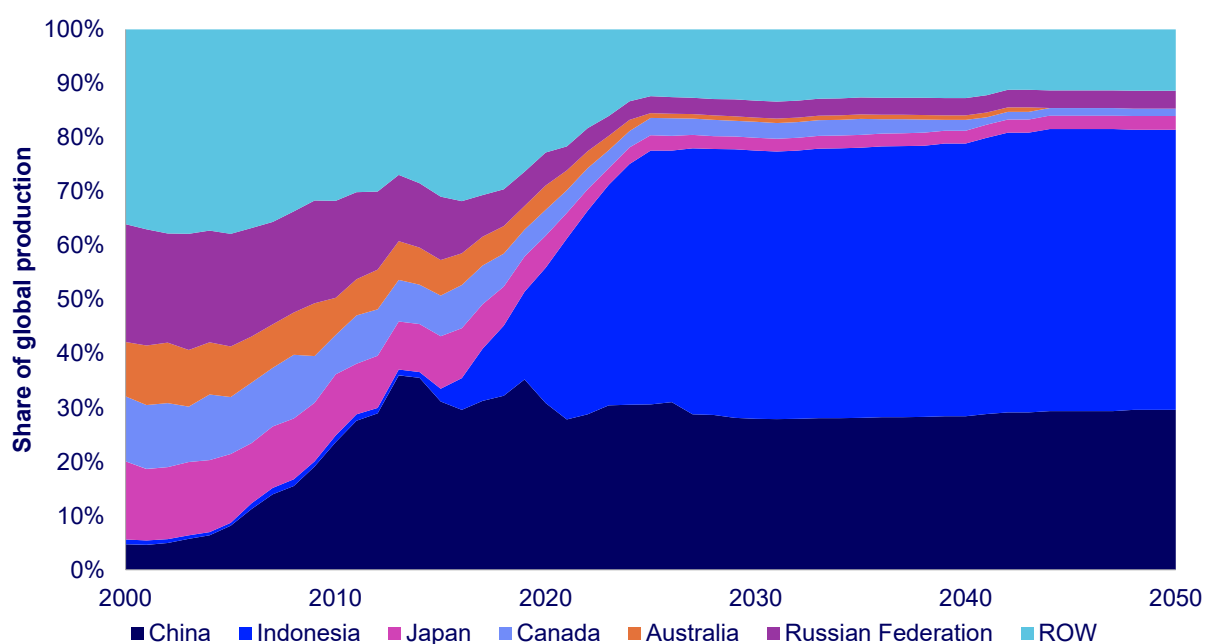
The primary tool at the government's disposal has been the implementation of stringent RKAB (Rencana Kerja dan Anggaran Biaya) - Indonesia's Mining Business Work Plan and Budget – mining quotas, changes to which have sent immediate ripples through markets since 2023. The quotas set out a volume that each miner is permitted to produce on a wet metric tonne (wmt) basis of nickel ore, with miners applying for quotas (with an option to revise mid-term). Notably, in 2025 the quota process reverted from a three-year approval term back to an annual term to tighten resource management, starting in 2026. Annual changes to quota allocations are ostensibly contingent on factors including market conditions and miners' environmental performance but remain ultimately down to the government's discretion.

For the nickel market, RKAB quotas have quickly become the most critical factor in the supply-side equation and are likely to define market dynamics (including the global balance) for at least the next decade. Indonesian officials have indicated they would periodically raise or lower RKAB quotas to support nickel prices within a range deemed sufficient for the domestic sector's profitability, while avoiding incentivising supply growth from other regions and/or resulting in possible demand destruction.

Major producing countries

In the early 2000s, China embarked on its vast programme of industrialisation and economic development, and for all its new infrastructure it needed stainless steel and, more importantly, nickel to produce the stainless. At that time, China was poorly endowed with domestic nickel production and the west saw this as a massive opportunity and began developing projects to capitalise on this growth. Ultimately, all these projects were delayed for various reasons and China was left wanting in terms of the nickel units it needed and had to go it alone. Out of adversity, NPI was born and China would never again look to rely on other nations to meet its aspirations.

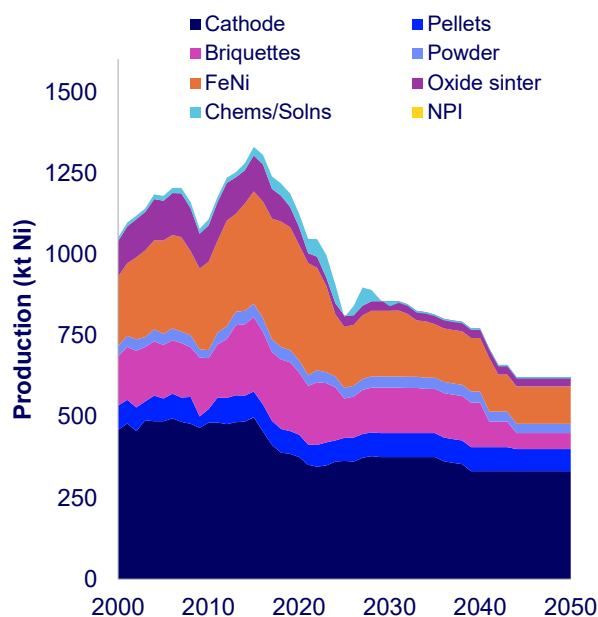
Refined production share by country



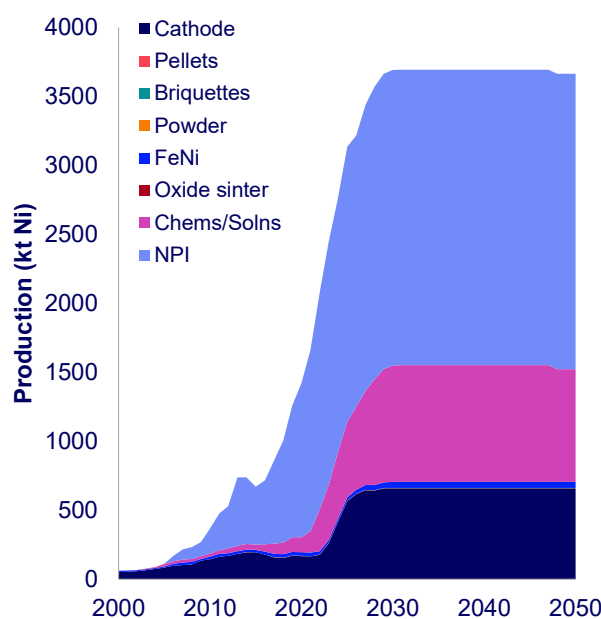
Source: Wood Mackenzie

This resulted in China accounting for 20% of global refined production by 2010 (up from 5% in 2000), fed by imports of nickel ores from, predominantly, the Philippines and then Indonesia. When Indonesia closed its doors to ore exports to force more domestic, higher value production, China once again grabbed the opportunity and by 2020, Indonesian production (which was Chinese controlled and primarily destined for the Chinese market) combined with the existing NPI facilities in China itself accounted for more than 50% of global output. This share stands at 78% in 2026.

Production outside Indonesia and China



Production Indonesia and China



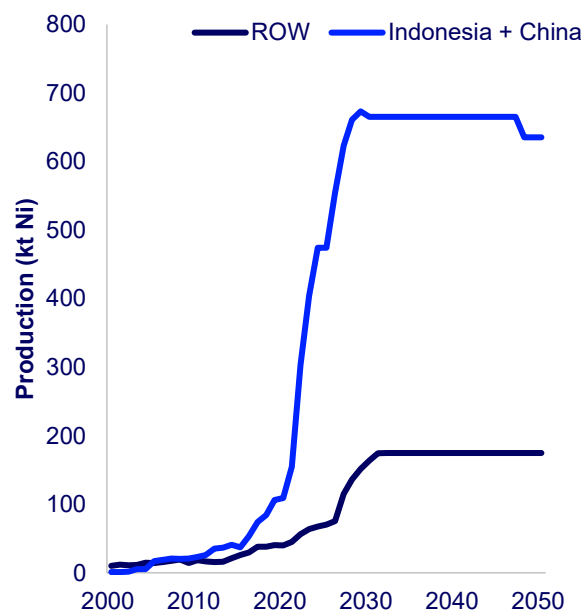
Over the past decade (2015 to 2025) refined production in China has increased by 83% (579 kt), although NPI production has fallen from 63% of total output to 25%, as the centre of gravity has shifted towards Indonesia and Chinese refiners have built chemical refining capacity to feed the battery industry. In Indonesia over the same period, production has jumped by 1.8 Mt, much of which has centred on NPI to feed the Sino-Chinese stainless-steel industry.

As noted above, the latest wave of developments (since 2020) is the advent of Chinese-backed HPAL projects in Indonesia to feed the growing battery industry, which have been remarkably successful in terms of ramping up to production. However, growth in demand for batteries has since slowed, in large part due to China's pivot towards iron-based LFP cathode chemistries. As a result, excess nickel sulphate is now being converted to nickel metal in China, much of which is Class I and acceptable for delivery at the LME. And what was perhaps the last bastion of western control in the nickel market – that of LME deliverable metal – is now also dominated by Indonesia and China.

Meanwhile, outside these two countries, nickel production has *fallen* by 34% (450 kt) between 2015-2025 as project economics have come under pressure from lower-cost competition in Indonesia. Australia is a notable victim of the rise of the Sino-Indonesian nickel sector, with output decreasing 74% (98 kt) during this period. Australian casualties include BHP's Nickel West complex, which had re-tooled one third of Kwinana refinery capacity to produce battery-grade nickel sulphate prior to being taken offline.

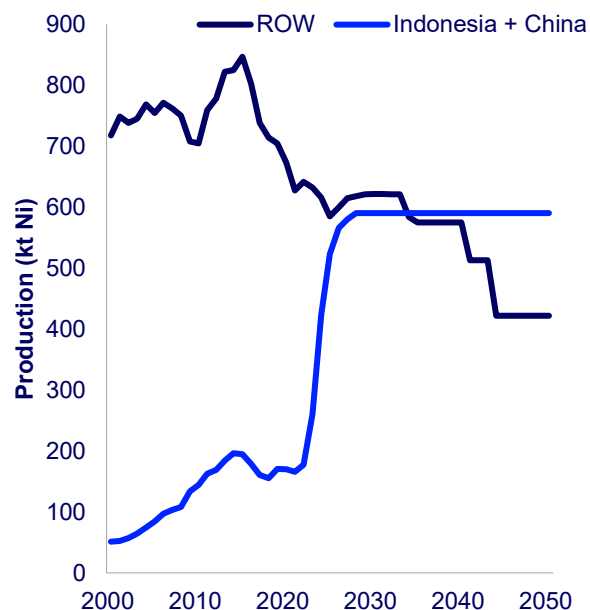
Looking forward to 2030 and 2040 and beyond, Sino-Indonesian dominance as nickel producers will increase to more than 80% of global output. This dynamic cannot be ignored and needs to be accepted and worked with rather than worked around.

Nickel sulphate production



Source: Wood Mackenzie

Class I production



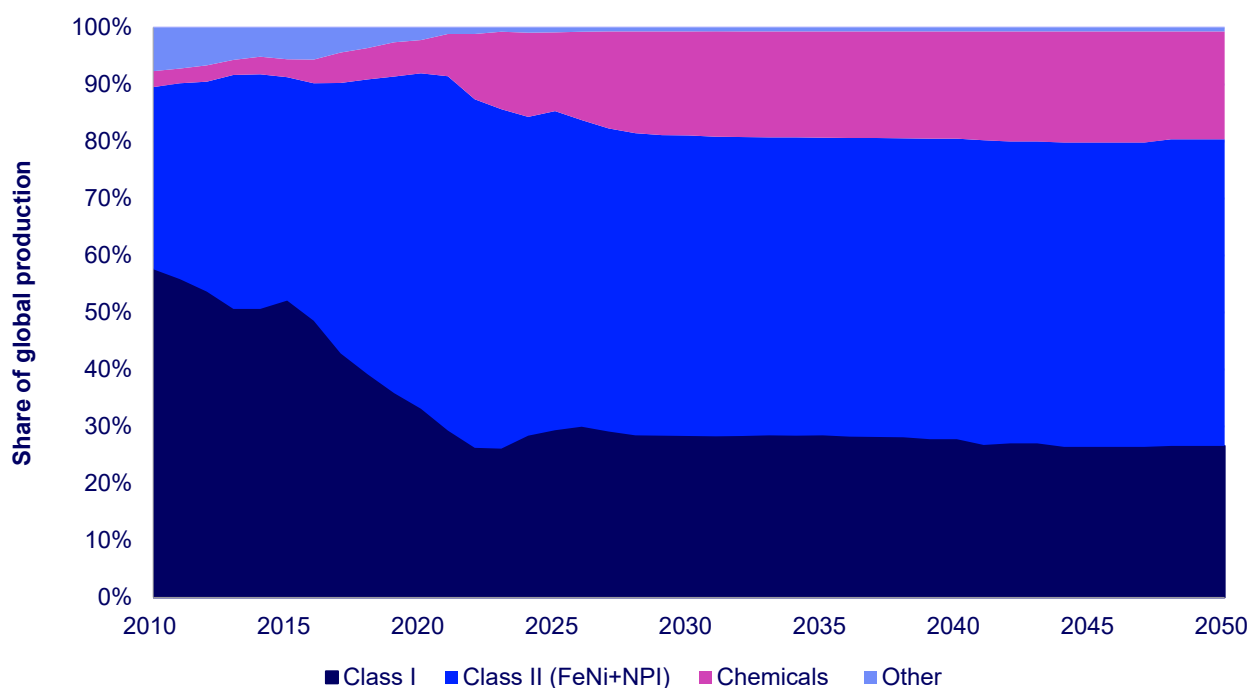
Source: Wood Mackenzie

Nickel product type and future benchmark pricing

A knock-on impact from the changing geographical production landscape has been a change in the dominant nickel products in the market and the “true” price of nickel.

For decades, Class I nickel products were the dominant form, but slipped from 70% of the market to under 60% by 2010, and to less than one third by 2020. However, the decline in market share was a result of the massive increase in NPI output rather than a decline in Class 1 output. With technological developments in the stainless steel sector which

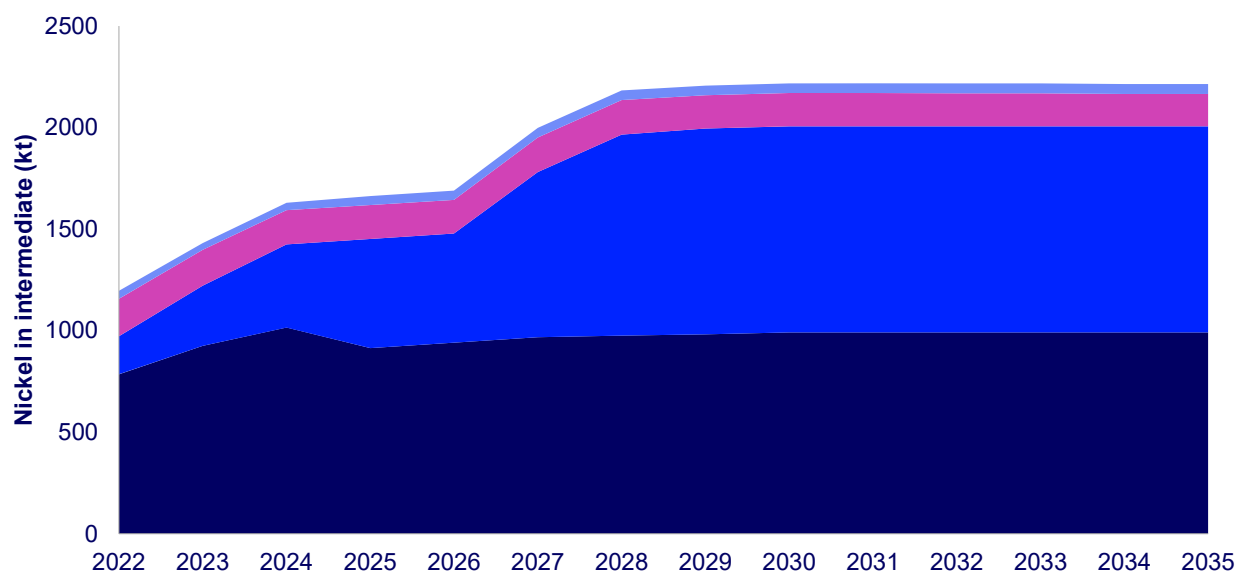
Nickel production by Class



uses other forms of nickel-containing products – from Class II FeNi to stainless scrap - there was no need to buy Class I metal to satisfy this demand. Indeed, the iron content of NPI, FeNi and scrap was also desirable for stainless steelmaking, while the free Cr content in NPI (approx. 4%) was also appreciated.

Nickel consumption for EV batteries was a booming demand area for Class I metal. Batteries need nickel sulphate, and the dissolution of Class I briquettes and pellets was a mainstay of nickel sulphate production. However, with the successful implementation of large tonnages of HPAL processing capacity and the re-purposing of NPI lines to produce matte (as well as the construction of dedicated matte facilities) in Indonesia, the most-used feed source for sulphate has now become nickel intermediates (MHP – mixed hydroxide precipitate – and nickel matte).

Smelter production by intermediate type



Source: Wood Mackenzie

NB: Sulphate in this chart is crude nickel sulphate by-product which needs further refining

■ Matte ■ MHP ■ MSP ■ Sulphate

The growth in capacity for intermediates production has been exponential. Global nickel-in-MHP output rose from 77 kt in 2020 to almost 540 kt in 2025 and is expected to remain at that level in 2026. However, the demand for nickel in EV batteries has slowed, resulting in an oversupply of intermediates and nickel sulphate. And this is having a big influence on intermediate payabilities, sulphate prices, and ultimately LME nickel prices.

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